For those planning to connect the USB Differential Pair to the STM32 :

Don’t forget to add an external crystal oscillator to the STM32 (HSE) : it’s mandatory to be able to use the USB PHY

Be aware of the layout constraint and considerations for high-speed differential signals : Document yourself on how to properly route USB diff pair with KiCAD ! (netclasses, trace spacing and width, stackup consideration, JLCPCB impedance calculator)

These values (in yellow) should be fine :

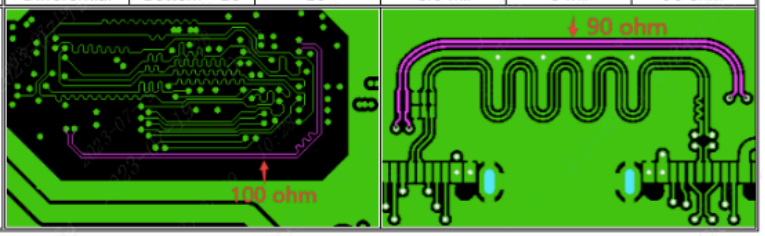
Impedance Calculation and Measurement of a PCB Track

For orders requiring impedance control, it is essential to provide your impedance requirements in the form of a table or diagram, along with the compressed PCB files. The general value of impedance of tracks are listed in the figure below with trace width, gap and layer information.

Table

Type	Layer	Reference	Trace Width	Trace Gap	Impedance
Single Ended	Top – L1	L2	6.2 mil	/	50 ohm
Differential	Top – L1	L2	5.6 mil	6 mil	90 ohm
Single Ended	Inner – L3	L2, L4	5.6 mil	/	50 ohm
Coplanar DP	Inner – L3	L2, L4	4.4 mil	8 mil	100 ohm
Single Ended	Bottom – L6	L5	6.2 mil	/	50 ohm
Differential	Bottom – L6	L5	5.6 mil	6 mil	90 ohm

Diagram



Source : https://jlcpcb.com/blog/beginners-guide-to-impedance-control-in-pcb